3.4.3 Comparision Between Magnetic and Electric Circuits

S.NO	PROPERTIES	MAGNETIC CIRCUIT	ELECTRIC CIRCUIT
1	Definition	The closed path followed by magnetic flux is called magnetic circuit.	The closed circuit followed by electric current is called electric circuit.
2	Driving Force	MMF is the pressure required to set up the magnetic flux in magnetic circuit (Ampere-Turn)	EMF is the pressure required to set up the current in an Electric circuit (Volt).
3	Response	FLUX $(\emptyset) = \frac{MMF}{Reluctance}(weber)$	CURRENT$(I) = \frac{EMF}{Resistance}(Ampere)$
4	Impendance	RELUCTANCE(S)=$l/(\mu_0\mu_r A)$ [AT/Weber]	RESISTANCE (R)=$\rho l/A$(ohms)
5	Admittance	PERMEANCE $= \frac{1}{Reluctance}[wb/AT]$	CONDUCTANCE $= \frac{1}{Resistance}(Siemens)$
6	Proportionality	RELUCTIVITY$= \frac{1}{Permeability}(M/H)$	RESISTIVITY$= \frac{1}{Conductivity}(ohm-meter)$
7	Density	FLUX DENSITY B = μ H(wb/m^2)	CURRENT DENSITY J=I/A($Amp/m^{2)}$
8	Field Intensity	MAGNETICFIELD INTENSITY(H)=NI/l(AT/m)	ELECTRIC FIELD INTENSITY=E/l(volt/m)

Whenever a conductor or coil is moved or rotated in a magnetic field and cut the magnetic line of force (flux), an EMF will be induced in that conductor or coil.

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3.5.1 Faraday's Law of Electromagnetic Induction

FIRST LAW: Whenever a conductor cuts magnetic flux, an EMF is induced in that conductor.

SECOND LAW: The magnitude of the induced EMF is directly proportional to the rate of change of flux linked with the conductor.

The two types of EMF induced are:

i) Dynamically induced EMF.
ii) Statically induced EMF.

3.5.2 Dynamically Induced EMF

Moving a coil/conductor in a uniform magnetic field will induce an EMF which is known as dynamically induced EMF. Generators work on this principle.

Consider a conductor of length *l* (meters) placed in a uniform magnetic field of density B(wb/m^2), moved with a velocity V(m/s) perpendicular to the direction of the

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